Evaluation

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Definition 0.1. (value)

$$F(x) :\simeq \Big(\text{if}((F \text{ family}) \wedge (x \in \text{dom}(F))) \text{ then } \bigcup\{s|(x,s) \in F\} \text{ else undef exception_else undef}\Big)$$

1 Value

Definition of the value of the function:

Definition 1.1. (semi-equality)

$$I_{\text{Fm}}(s \approx t) : \Longleftrightarrow$$

Definition 1.2. (equality)?

Definition 1.3. (quasi-equality)(Note: it is good for semantics with partial functions, not the two level set-semantics i use.)

$$(s \simeq t) : \Longleftrightarrow \ (((s \text{ defined}) \vee (t \text{ defined})) \ \longrightarrow \ (s = t))$$

Definition 1.4. (underdefined class)(Note: be careful, since not every undefined class is underdefined, some are overdefined in various ways, so not even semantically equal to eachother.)

$$\text{undef} :\simeq (\varepsilon x : \bot)$$

Definition 1.5. (if-then-else)todo

$$(\text{if } \varphi \text{ then } t_1 \text{ else } t_2) :\simeq (\{z|((z \in t_1) \wedge \varphi) \vee ((z \in t_2) \wedge \neg\varphi)\})$$

Side conditions:
(z is not free in t_1),
(z is not free in t_2),
(z is not free in φ).

What if φ is assigned with U (i.e. it is assigned with the "uncertain" truth value)?
If $(I(\varphi) = U)$, then

$$(I((\text{if } \varphi \text{ then } t_1 \text{ else } t_2)) = I(\text{undef}))$$

should follow.
Btw, $I(\text{undef}) = \emptyset$.

2 Some simple theorems about classes

Definition 2.1. (some syntactic sugar)

$$(x \text{ any}) : \longleftrightarrow \top$$

Context: (x:Var)

Definition 2.2. (Semi-singleton)

$$(s \text{ ss}) : \longleftrightarrow (!y : y \in s)$$

Side condition:
Variable y is not in free variables of term s.

Definition 2.3. (Semi-singleton subset relation)

$$(s \text{ sss } t) : \longleftrightarrow ((s \text{ ss}) \wedge (s \subseteq t))$$

Definition 2.4. (Function constructor)

$$(\lambda x : t) := \{(x, t) | (x \text{ any})\}$$

Context: (x:Var) (t:Term)
Comment: Term t can contain x as a free variable.
Side conditions:
(variable x is bound by class collection in r.h.s)
(variable x is bound by lambda abstraction in l.h.s)

Definition 2.5. (Family constructor)

$$(\Lambda x : t) := \{(x, s) | (s \text{ sss } t), x \text{ any}\}$$

Term t can contain x as a free variable.
Side conditions:
Variables x and s are different.
Variable s is not in free variables of t.
As term s can contain x as a free variable.

Definition. (LOCAL definition of a value of a function, value of a family, for the next theorem)

$$f(x) := \bigcup \{s | (x, s) \in f\}$$

Theorem 2.1. (Evaluation of a function) Context:
(x:Var)(t:Term)(z:Term)

$$(z \text{ set}) \wedge (t[z/x] \text{ set}) \longrightarrow$$

$$(\lambda x : t)(z) = t[z/x]$$

Proof.
To prove the equality, we use set extensionality and MP.

$$((\lambda x : t)(z) \subseteq t[z/x]) \wedge ((\lambda x : t)(z) \supseteq t[z/x]) \longrightarrow ((\lambda x : t)(z) = t[z/x])$$

($\subseteq$)
Assume

$$q \in (\lambda x : t)(z)$$

By definition of $-(-)$ (evaluation),

$$q \in \bigcup \{b | (z, b) \in (\lambda x : t)\}.$$

By destruction of the union,

$$\exists s \in \{b | (z, b) \in (\lambda x : t)\} : q \in s.$$

Note for the following step: since variable b has no free occurences in t, $t[s/b] = t$.

$$s \in \{b | (z, b) \in (\lambda x : t)\} \implies (s \text{ set}) \& ((z, s) \in (\lambda x : t))$$

By definition of λ,

$$(z, s) \in (\lambda x : t) \implies (z, s) \in \{(x, t) | \top\}$$

Unfolding the notation:

$$(z, s) \in \{(x, t) | \top\} \implies (z, s) \in \{w | \exists x : w = (x, t) \wedge \top\}$$

$$(z, s) \in \{w | \exists x : w = (x, t) \wedge \top\} \implies \exists x : (z, s) = (x, t) \wedge \top$$

Notes: w is free in t, w is different from x. Since otherwise we would write $(x[(z, s)/w], t[(z, s)/w])$.
Take such x. By defining property of ordered pairs,

$$(z, s) = (x, t) \implies (z = x) \& (s = t)$$

$$(q \in s) \& (s = t) \implies q \in t$$

$$z = x \implies t[z/x] = t$$

Therefore

$$(q \in t) \& (t[z/x] = t) \implies (q \in t[z/x])$$

($\supseteq$)
Assume $(q \in t[z/x])$.
Let $s := t[z/x]$. Then $(q \in s)$.
Goal1:$(z, s) \in (\lambda x : t)$.

$$x[z/x] = z$$

$$(z, t[z/x]) = (x[z/x], t[z/x]) = (x, t)[z/x]$$

Goal1: $(x,t)[z/x] \in \{(x,t)|(x \text{ any})\}$
Goal2: $\exists x : (x,t) = ((x,t)[z/x])$
x:=z
There are no free occurencies of x in $((x,t)[z/x])$, because of the substitution.
$c := ((x,t)[z/x])$

$$(((x,t)[z/x]) = c) \longrightarrow \exists x : (x,t) = c$$

Goal' is done.

From the premise, $(x,t)[z/x]$ is a set.
Therefore Goal1 is proven.
So $(x,t)[z/x] \in (\lambda x : t)$.
So $(z, t[z/x]) \in (\lambda x : t)$.
So $t[z/x] \in \{b|(z,b) \in (\lambda x : t)\}$.
Since $(q \in t[z/x])$, we also have

$$q \in \bigcup \{b|(z,b) \in (\lambda x : t)\}$$

So $q \in (\lambda x : t)(z)$.
And we are done.
Done.

Theorem 2.2. (Evaluation of family)

$$(\Lambda x : t)(z) = t[z/x],$$

and in particular:

$$(\Lambda x : t)(x) = t.$$

Proof.
By definitions,

$$(\Lambda x : t)(z) = .$$

Done.

2.1 Lambda operatorOLD

Definition 2.6. not good anymore: doesn't work for classes Lambda operator.

$$(\lambda x : t) = \{p|\exists x : p = \langle x, t\rangle\}$$